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Part 1 Random walk Metropolis

In the first task the Random Walk Metropolis algorithm was applied to draw samples from the double-exponential (Laplace) target distribution:

$$f(x) = (1/2) \exp(-|x|)$$

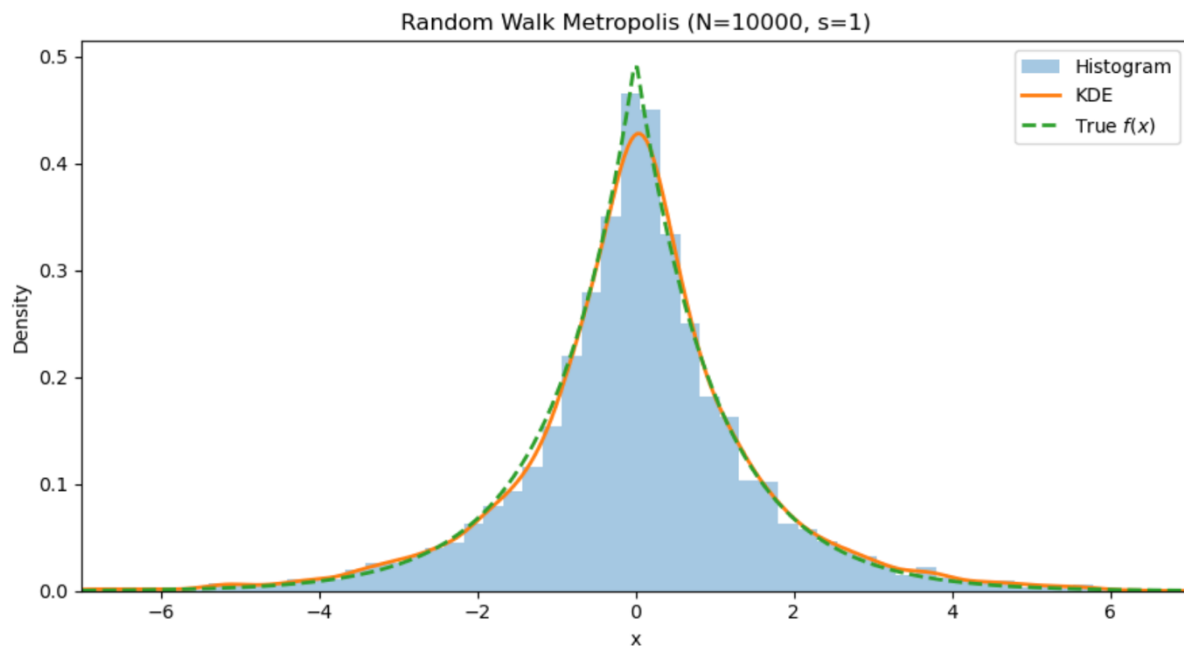
This distribution has mean 0 and standard deviation $\sqrt{2} \approx 1.4142$.

Part 1(a)

The algorithm was initialised at $x_0 = 0$ with $N = 10,000$ iterations and standard deviation $s = 1.0$ according to the task.

At each step a candidate x^* is drawn from $N(x_{i-1}, s)$ and accepted with probability $\min(1, f(x^*)/f(x_{i-1}))$. To prevent avoid numerical errors I used the log-ratio criterion: the candidate is accepted whenever $\log u < \log f(x^*) - \log f(x_{i-1})$.

Statistic	Estimate (python)	Estimate (R)	True
Mean	0.019	-0.000569	1
Standart Deviation	1.49	1.347895	$\sqrt{2} \approx 1.4142$



The Monte Carlo estimates are close to the true values, confirming that the chain has explored the target distribution effectively. The histogram and KDE closely track the true $f(x)$

curve across the range $[-7, 7]$, with the KDE providing a smooth continuous approximation that overlaps well with the theoretical density.

Part 1(b)

Convergence was assessed using the R-hat statistic. Four chains were run with different initial values of 2, -1, +1, +2. The R-hat statistic compares within-chain variance (W) to between-chain variance (B):

$$\text{R-hat} = \sqrt{(B + W) / W}$$

Values of R-hat near 1.0 indicate convergence; the conventional threshold is $\text{R-hat} < 1.05$.

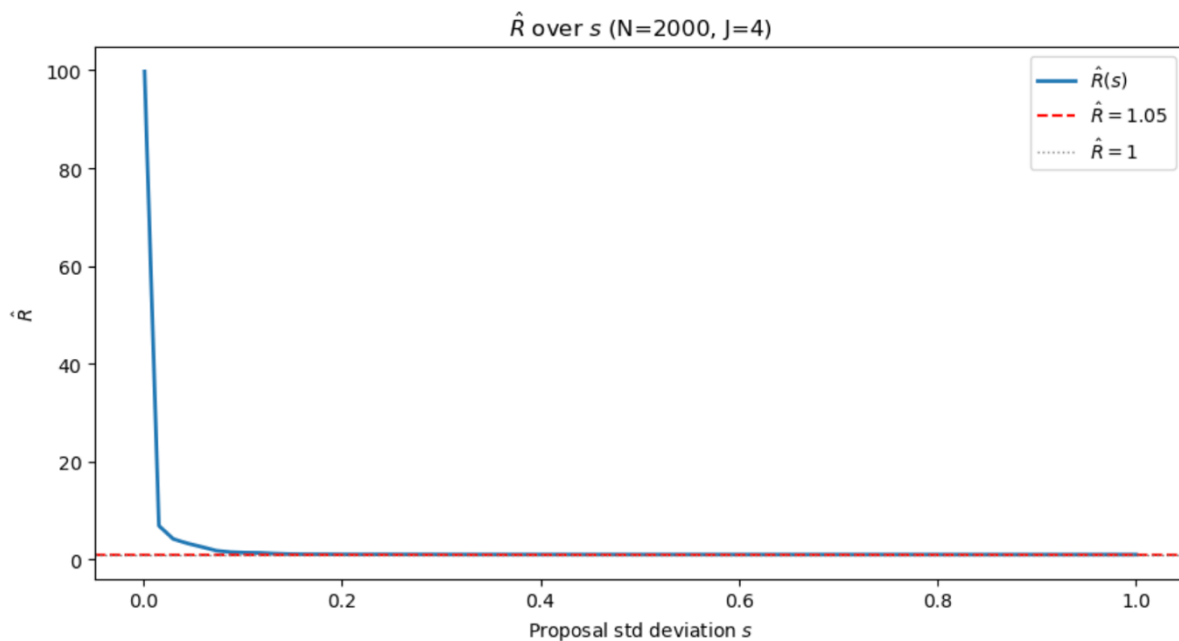
Results for $s = 0.001$:

R-hat (Python) = 99.75 and R-hat (R) = 103.14 far above the 1.05 threshold. With a very small proposal standard deviation the chains move extremely slowly and each chain stays near its starting value throughout the 2,000 iterations, so the between-chain variance vastly exceeds the within-chain variance.

R-hat across $s \in [0.001, 1.0]$:

R-hat < 1.05 for $s \in [0.204, 1.000]$ (Python)

R-hat < 1.05 for $s \in [0.291, 1.000]$ (R)



The results indicate that $s \geq 0.2$ is required for the random walk Metropolis algorithm to adequately explore the Laplace target within $N = 2,000$ steps.

Part 2: Flight Data Analysis (2004–2008)

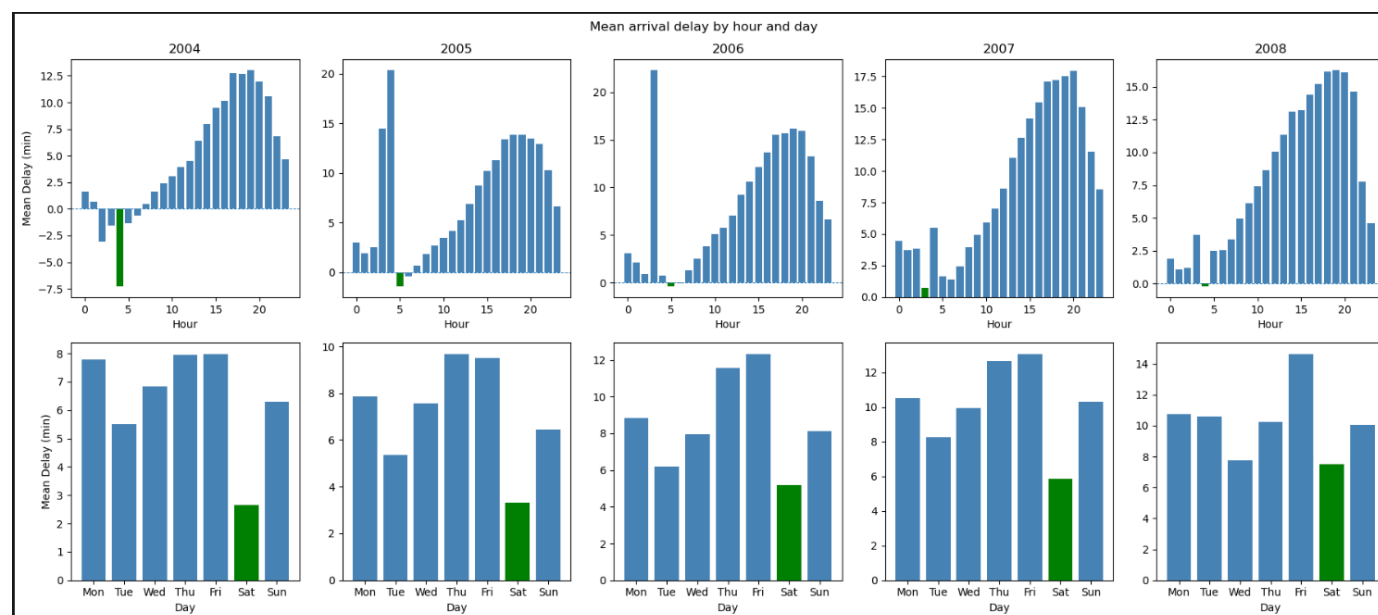
A subset 2004–2008 was selected, covering approximately 30.6 million non-cancelled flight records across five annual CSV files. Three supplementary files were used in the research: plane-data.csv (5,029) for manufacture year, airports.csv (3,376) for geographic coordinates.

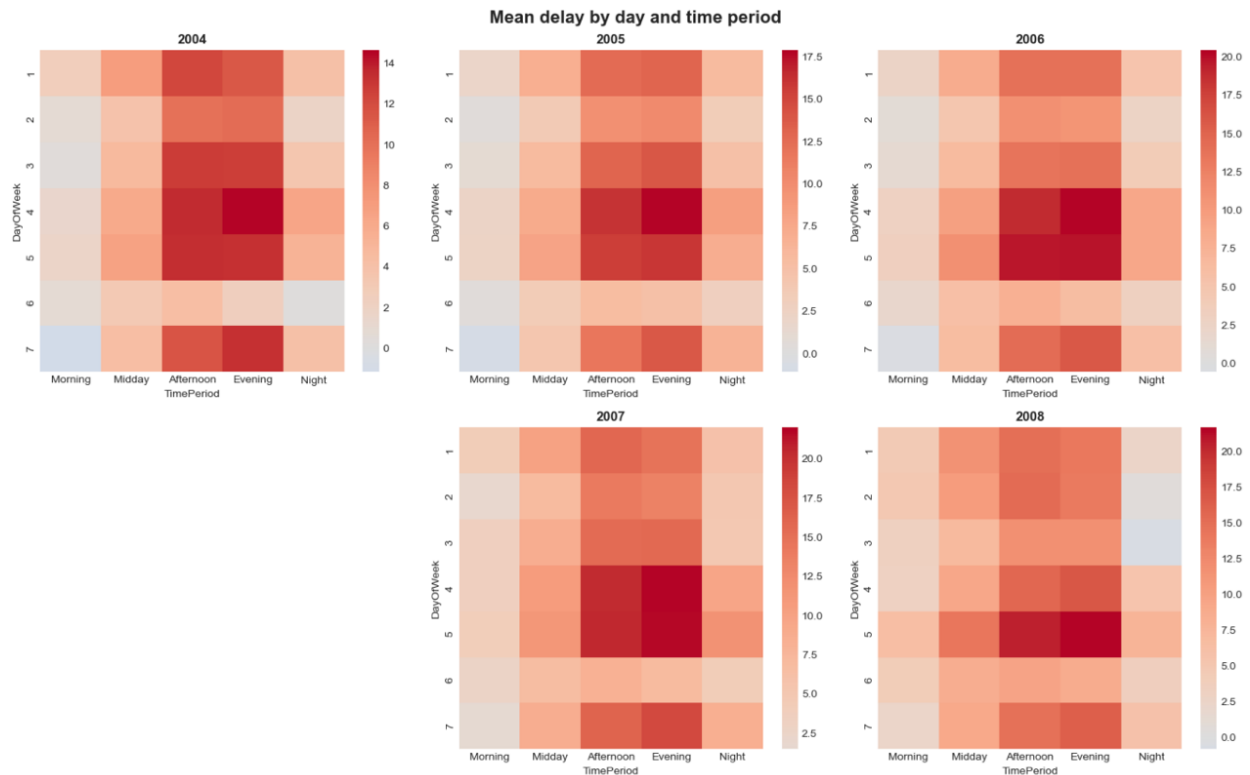
Data wrangling steps:

1. Rows with Cancelled = 1 were removed. Only non-cancelled flights contribute to delay and diversion analyses, as cancelled flights have no recorded ArrDelay.
2. ArrDelay, DepDelay, Distance, CRSDepTime, CRSArrTime and Diverted were cast to numeric, with non-parseable values set to NA.
3. DepHour = CRSDepTime / 100 extracts the scheduled hour (0–23) from the four-digit HHMM field.
4. UniqueCarrier was integer-encoded, enabling its use as a numeric feature in logistic regression.
5. Origin and Dest IATA codes were joined to airports.csv to attach latitude and longitude for both departure and arrival airports (OriginLat, OriginLong, DestLat, DestLong).
6. After merging with plane-data.csv on TailNum, PlaneAge = Year – MYear. Negative ages and ages above 50 were excluded as data quality filters. Age groups with fewer than 50 observations were also excluded.

Part (a) What are the best times and days of the week to minimise delays each year?

For each year, mean arrival delay (ArrDelay) was aggregated by scheduled departure hour (0–23, derived from CRSDepTime) and day of week (1=Monday ... 7=Sunday). Bar charts (by year) and heatmap (by time period) were generated to visualise patterns.





The results are consistent across all five years and across both implementations. Early morning departures (03:00-05:00) consistently achieve the lowest mean arrival delays, often resulting in early arrivals. Aircraft operating the first departure of the day begin with no delays from earlier rotations, minimal gate congestion, and low air traffic control workload. As the day progresses, knock-on delays accumulate across the network, so evening slots (18:00–21:00) carry the highest average delay.

Saturday is the best day to fly in every year studied. Business travel demand is lowest on Saturdays, meaning fewer passengers, shorter queues, and less gate congestion. Thursday and Friday are consistently the worst days, coinciding with peak business travel and leisure travel overlap.

The result of the analysis are the following:

2004: best hour = 4:00, best day = Sat

2005: best hour = 5:00, best day = Sat

2006: best hour = 5:00, best day = Sat

2007: best hour = 3:00, best day = Sat

2008: best hour = 4:00, best day = Sat

Part 2 (b) Evaluate whether older planes suffer more delays on a year-to-year basis.

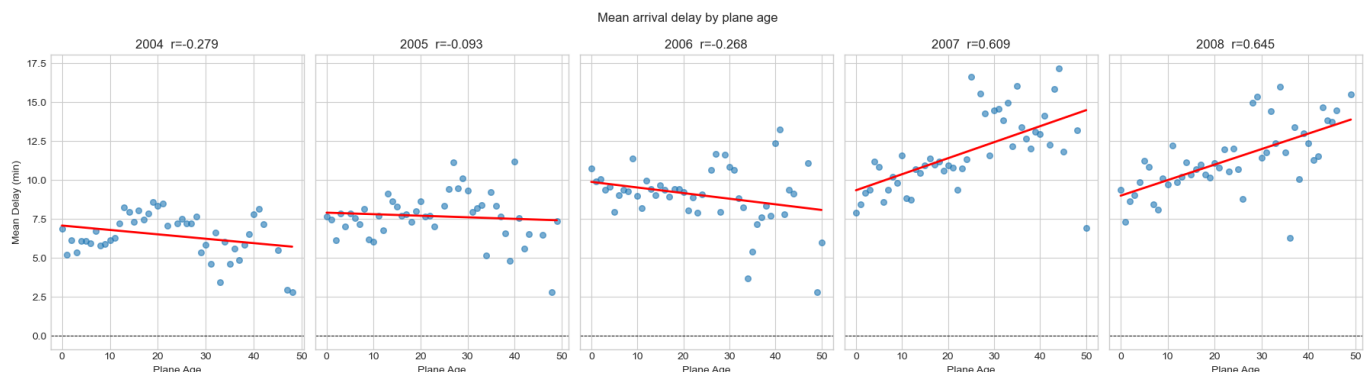
To assess the relationship between aircraft age and arrival delay, flight records were merged with Plane-data.csv on TailNum. Plane age was computed as $\text{PlaneAge} = \text{flight year} - \text{manufacture year}$. Records with $\text{PlaneAge} < 0$ or $\text{PlaneAge} > 50$ were excluded as likely data errors. Mean arrival delay was then computed at the (Year, PlaneAge) level, and age groups with fewer than 50 observations were dropped to avoid unreliable cell estimates.

Statistical analysis Pearson correlation:

Pearson's r and its p -value were computed between the PlaneAge index and the mean ArrDelay for each age group.

2004–2006 (negative / near-zero r , $p > 0.05$): There is no statistically significant evidence that older planes suffer more delays in these years. The slightly negative correlations suggest that in this period, aircraft age did not translate into greater delay risk - possibly because older planes were concentrated on mature, well-managed routes with predictable scheduling, while newer aircraft were being deployed on more demanding or expanding routes subject to greater operational variability. Fleet composition and maintenance standards may also have offset any age effect.

2007–2008 (positive $r \approx 0.61$ – 0.65 , $p < 0.001$): A statistically significant and moderately strong positive correlation emerges: older planes are associated with higher mean arrival delays. By 2008 the relationship strengthened further ($r = 0.645$), consistent with continued deterioration of older fleet performance under financial strain.



The evidence for older planes suffering more delays is mixed but leans positive over the five-year window. The relationship is absent in 2004–2006 and significant in 2007–2008.

Part 2(c) logistic regression model for the probability of diverted US flights

For each year a logistic regression model was fitted to predict the binary outcome *Diverted* (1 = flight diverted to an unplanned airport, 0 = not diverted). Diversion is a rare event in this dataset, so without correction a naive classifier would simply predict 0 for every flight. Class-balanced sample weights were therefore applied in both implementations to ensure that diverted lights contribute meaningfully to the model.

Year Diversion rate

2004 0.197%

2005 0.200%

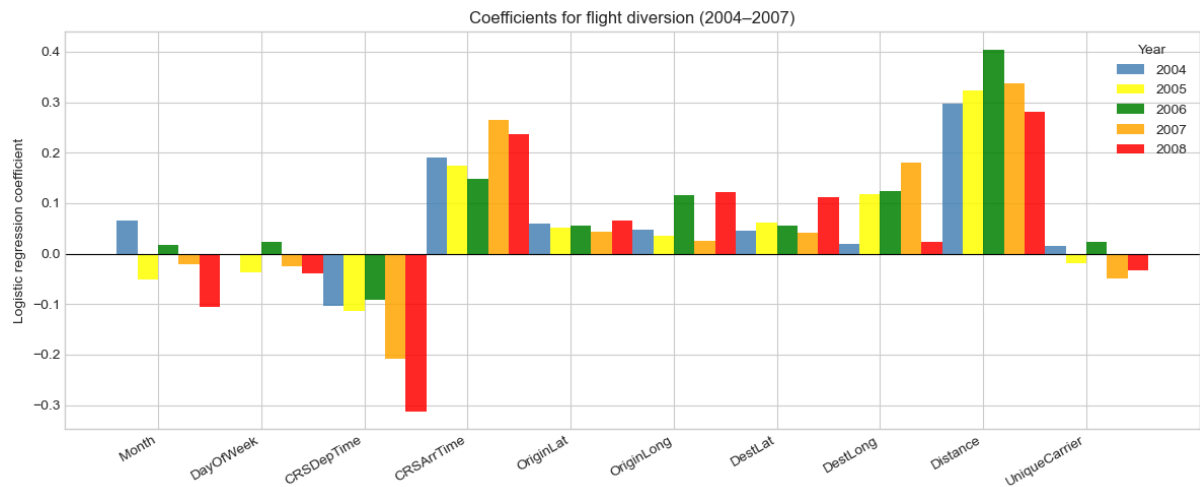
2006 0.231%

2007 0.236%

2008 0.243%

Diversion rates rose over the five years suggesting a modest but consistent deterioration in the ability of airlines to complete flights as planned.

Coefficient interpretation:



All features were standardised before fitting, so coefficients are directly comparable in magnitude. The bar chart shows the following patterns:

Distance is the dominant predictor of flight diversion across all years from 2004 to 2008, with standardized coefficients ranging from 0.28 to 0.40. Later scheduled arrival times consistently increase diversion risk (coefficients 0.15–0.27).

A striking trend emerges for scheduled departure time: while the effect was weakly negative from 2004–2006 (–0.09 to –0.11), it strengthened substantially to –0.21 in 2007 and –0.31 in 2008. This suggests that later departures became progressively less likely to result in diversion over the study period.

Month, day of week, and airport coordinates show weak and inconsistent effects, providing little predictive value beyond distance and timing variables. The carrier variable as encoded shows near-zero coefficients, but this reflects encoding limitations rather than true absence of carrier effects.